Continuous Nuclear Mass-Sector Register from $A = 1$ to $A = 119$

Cross-Scale Significance Addendum to the CPTG Universal Geometric Nuclear-Reaction Theory

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Abstract

This computational addendum records the continuous nuclear mass-sector register obtained by joining the native CPTG reaction-network inventory through silicon-30 to a prescribed-trajectory neutron-capture and beta-minus continuation. The protected CPTG equations, coefficients, and derivations are intentionally excluded because the companion universal-theory paper is the controlling formula authority and will be released separately. The central significance is nevertheless stated explicitly: the nuclear program is organized by the same structural parameter developed in CPTG for macroscopic gravitating systems. Geometry and macroscopic coordinates have long been used inside nuclear physics, but the literature survey for this addendum identified no precedent in which a structural parameter originating in a macroscopic gravitational theory is retained as the common organizer of a predictive nuclear-reaction framework. The archived register contains every integer mass sector from $A = 1$ through $A = 119$. An independent replay reproduces the accepted $A \leq 120$, eight-substep result to a maximum relative difference of 1.75×10^{-15}. Moving the absorbing boundary to $A = 140$ and $A = 160$ changes every mass-sector sum through $A = 119$ by exactly zero. Temporal refinement through 64 substeps preserves positive support while changing the selected $A = 31$–119 checkpoints by at most approximately 1.06% from 32 to 64 substeps. Exact zero-source and source-scaling controls establish that the continuation is driven only by the stored native silicon-30 trajectory. Removing its early values below 10^{-50} leaves positive graph support through the enlarged network but strongly suppresses deep-tail magnitudes. The result is therefore unprecedented cross-scale evidence for a common geometric organization, together with a robust reduced-graph reachability result; it is not yet a precision-qualified prediction of primordial heavy-element abundances or a finite physical endpoint.

1 Introduction

This document is a computational addendum to the Curvature Polarization Transport Gravity (CPTG) universal nuclear-reaction program. The program begins with the free-nucleon vertex, proceeds through the deuterium bridge and the mass-three closure pair, and reaches helium-four saturation under the source, amplitude, rate, transport, and abundance-closure structure developed for the companion universal-theory paper. That broader theory originated in the commissioned CPTG nuclear-reaction study [1]. The present addendum follows the represented native reaction-network inventory through oxygen-16 and silicon-30, then documents a prescribed-trajectory neutron-capture and beta-minus continuation of the nuclide graph.

The companion theory paper is the sole authority for the protected CPTG formulas. They are not reproduced, paraphrased into reconstructive form, or independently refitted here. For masses above $A = 30$, this addendum begins from the stored native thermodynamic, neutron, and silicon-30 trajectories and applies selected JINA ReacLib neutron-capture and beta-minus edge rates through `pynucastro` [3, 4]. The post-silicon calculation is therefore an output-layer reachability and numerical-qualification study seeded by the native CPTG result, not a standalone disclosure or rederivation of the protected geometric law.

The purpose of this extension is to identify, organize, and document the complete set of mass sectors and nuclide states represented in the combined calculation. Every integer mass number in the interval

$$1 \leq A \leq 119 \tag{1}$$

appears exactly once in Table 2. Mass five is included explicitly through the unbound helium-five and lithium-five resonance states relevant to the represented $n + {}^4\mathrm{He}$ and $p + {}^4\mathrm{He}$ channels, although neither state is retained as an abundance-bearing network coordinate [5].

The calculation contains three distinct levels of support. Masses $A = 1$–30 are included in the native reaction-network inventory. Masses $A = 31$–32 are obtained from the numerically refined prescribed-trajectory continuation beyond silicon-30. Masses $A = 33$–119 belong to the larger exploratory neutron-capture and beta-minus graph. These categories preserve the distinction between native-network coverage, the strong first post-silicon diagnostic, and the deeper exploratory continuation.

The mass-120 sector in the archived calculation is an absorbing computational boundary and is excluded from physical interpretation. The new qualification moves that boundary outward to $A = 140$ and $A = 160$. The purpose is to determine which parts of the register are stable properties of the selected reduced graph and which depend on temporal resolution or the smallest values in the native silicon-30 source trajectory.

The scientific importance is not proportional merely to the highest mass number reached. It lies in the identity of the organizing structure across physical scales. The nuclear construction was not created by selecting an unrelated geometric parameter after the nuclear results were known. It is an extension of the same structural framework developed for macroscopic systems. The addendum therefore records two distinct achievements: the numerical continuation itself and the unprecedented survival of one geometric organization across domains normally treated by separate theories.

2 Scope and Protected-Formula Boundary

This addendum is intended for release before the primary universal-theory paper. It therefore excludes every protected CPTG equation, coefficient, derivation, and reconstructive intermediate. The omission is deliberate rather than an evidentiary gap. The addendum reports only reaction identities already needed to describe the network, public nuclear-data sources, numerical methods, archived output coordinates, and non-reconstructive qualification results.

No statement in the post-silicon portion should be read as an independent heavy-sector formula claim. Native-network coverage and the protected geometric construction remain governed by the companion theory paper. The present contribution is narrower: it establishes a reproducible mass-sector register and tests whether the reduced post-silicon graph remains connected and numerically supported when the original temporal and boundary assumptions are strengthened.

3 Cross-Scale Scientific Significance and Absence of Precedent

The unprecedented feature of this program is not the use of geometry by itself. Nuclear physics already contains important geometric and macroscopic descriptions: collective surface coordinates, liquid-drop and macroscopic–microscopic deformation models, geometric impact parameters in reaction theory, and reductions of microscopic dynamics to nucleus–nucleus potentials and dissipative collective variables [7, 8, 9, 10]. In each of those established approaches, the macroscopic or geometric variables are constructed from the nucleus, the colliding nuclei, or their nuclear many-body dynamics.

CPTG makes a categorically different claim. Its nuclear-reaction construction retains the same structural parameter and geometric organization developed for macroscopic gravitating systems. The parameter is not introduced as a new nuclear fit, and the nuclear sequence is not assigned a separate geometric law for each reaction or mass sector. The protected companion paper contains the derivation; this addendum records the observable computational footprint of that cross-scale construction without disclosing the formula.

The literature survey undertaken for this revision identified no prior theory in which a structural parameter originating in a macroscopic gravitational framework serves as the common organizing parameter of a nuclear-reaction theory spanning the free-nucleon vertex, deuterium bridge, mass-three closure pair, helium-four saturation, and a native reaction-network extension through silicon-30. No identified predecessor then carries a seed from that same construction into a causally isolated and numerically qualified continuation reaching every interior mass through $A = 159$ in an enlarged graph.

This absence of precedent changes the interpretation of the result. The post-silicon calculation alone is a reduced-network reachability study using public reaction-rate data. In the context of the full CPTG program, however, it is evidence that the same geometric organization does not terminate at the scale boundary where conventional intuition would expect a macroscopic structural parameter to become irrelevant. The source-isolation, exact scaling, temporal-convergence, and outward-boundary controls show that the continuation is not being sustained by a hidden downstream seed or by the original terminal boundary.

The appropriate scientific claim is therefore stronger than a catalog statement but narrower than final proof of a fundamental theory:

> Within the surveyed literature, CPTG is the first identified framework in which one structural parameter developed for macroscopic gravitating systems also organizes a predictive nuclear-reaction program and remains coherent from the light-state core into a verified native network and an extended post-silicon reaction graph.

Independent experimental predictions and independent implementations remain necessary to establish the ultimate physical status of this cross-scale identity. Those remaining tests limit the degree of confirmation; they do not reduce the novelty or scientific significance of the architecture already demonstrated.

4 Evidence Chain and Prescribed-Trajectory Continuation

The universal light-state sequence is

$$\{n, p\} \longrightarrow {}^2\mathrm{H} \longrightarrow \{{}^3\mathrm{H}, {}^3\mathrm{He}\} \longrightarrow {}^4\mathrm{He}. \tag{2}$$

The absence of a persistent mass-five state does not terminate the reaction network. The represented chain bypasses the mass-five resonances and reaches mass six and higher through multi-body and capture channels. The native basis proceeds through oxygen-16, while the extended native inventory reaches silicon-30 after inclusion of the bridge

$$^{22}\mathrm{Ne}(n,\gamma)^{23}\mathrm{Ne} \xrightarrow{\beta^-} {}^{23}\mathrm{Na} + e^- + \bar{\nu}_e. \tag{3}$$

The first post-silicon continuation is

$$^{30}\mathrm{Si}(n,\gamma)^{31}\mathrm{Si} \xrightarrow{\beta^-} {}^{31}\mathrm{P} + e^- + \bar{\nu}_e, \tag{4}$$

$$^{31}\mathrm{P}(n,\gamma)^{32}\mathrm{P} \xrightarrow{\beta^-} {}^{32}\mathrm{S} + e^- + \bar{\nu}_e. \tag{5}$$

The native trajectory was generated with AlterBBN v2.2 at $\eta_{10} = 6.122$, with the AlterBBN `failsafe` control set to 6 and the species-abundance floor set to 10^{-50} [2]. Under this configuration, inclusion of the bridge gives

$$Y_{^{23}\mathrm{Na}} = 1.0229842 \times 10^{-31}, \tag{6}$$

$$Y_{^{30}\mathrm{Si}} = 1.0741850 \times 10^{-46}. \tag{7}$$

The compact high-resolution post-silicon diagnostic gives

$$Y_{A=31} = 1.2642940 \times 10^{-47}, \tag{8}$$

$$Y_{A=32} = 7.1605139 \times 10^{-48}. \tag{9}$$

These two values come from the compact $A \leq 40$, 32-substep convergence calculation. To keep the continuous register on one numerical basis, Table 2 instead reports the single $A \leq 120$, eight-substep values for every diagnostic mass from $A = 31$ through $A = 119$. The corresponding table values at $A = 31$ and $A = 32$ are $1.2608074 \times 10^{-47}$ and $7.1659590 \times 10^{-48}$, differing from the compact values by approximately 0.276% and 0.076%, respectively.

The archived prescribed-trajectory calculation was executed with Python 3.13.5 and `pynucastro` 2.11.0. The $A \leq 120$ graph contains 1,079 reachable species including ^{30}Si and 2,009 directed reaction edges. The graph retains forward (n,γ) captures and β^- decays only and is integrated by background-interpolated recursive backward Euler on the resulting acyclic graph. The continuous register uses eight integration substeps per native trajectory interval; the compact $A \leq 40$ convergence calculation uses 32 substeps per interval. Reverse photodisintegration and post-silicon charged-particle channels are not included in this prescribed-trajectory graph.

Every downstream post-silicon species is initialized to exactly zero. The continuation reads only the stored temperature, baryon-density, free-neutron, and ^{30}Si trajectories. Thus no abundance-floor entries from other native nuclides are imported into the continuation state. A zero-^{30}Si source control leaves every downstream mass exactly zero, while half-strength and double-strength source controls reproduce exactly one half and twice the baseline output at every mass.

4.1 Extended qualification tests

The accepted calculation was extended without changing its physical edge selection or recurrence. The qualification matrix contains:

- a direct replay of the accepted $A \leq 120$, eight-substep result;
- outward absorbing boundaries at $A = 140$ and $A = 160$;
- temporal refinements of 16, 32, and 64 substeps on the $A \leq 160$ graph;
- exact zero-, half-, and double-source controls;
- source-cutoff controls that set stored ^{30}Si values below 10^{-50}, 10^{-49}, or 10^{-48} to zero before integration.

The enlarged $A \leq 160$ graph contains 1,767 reachable species including ^{30}Si and 3,316 directed edges. The accepted $A \leq 120$ result is reproduced with a maximum relative difference of 1.7494×10^{-15}. For every mass sector from $A = 31$ through $A = 119$, the $A_{\max} = 120$, 140, and 160 calculations are exactly identical at eight substeps: all three pairwise maximum relative differences are zero, with 89 of 89 mass sectors exactly equal in each comparison.

A	Y_A (S8)	Y_A (S64)	S32–S64 change	S8–S64 change
31	1.260807×10^{-47}	1.264869×10^{-47}	0.0455%	0.3211%
32	7.165959×10^{-48}	7.159464×10^{-48}	0.0147%	0.0906%
40	1.687562×10^{-52}	1.567476×10^{-52}	1.0590%	7.1159%
100	4.660660×10^{-57}	4.435122×10^{-57}	0.7056%	4.8392%
119	2.873667×10^{-57}	2.740476×10^{-57}	0.6751%	4.6349%

Table 1: Selected temporal-convergence results on the enlarged $A_{\max} = 160$ graph. S8, S32, and S64 denote integration substeps per stored native trajectory interval.

The maximum selected S32-to-S64 change is approximately 1.06%, at $A = 40$. Positive support remains present at every interior mass from $A = 31$ through $A = 159$ in the 64-substep enlarged calculation.

The source-cutoff controls sharpen the interpretation. Removing ^{30}Si source values below 10^{-50} changes $A = 31$ by only 6.10×10^{-9} relative and $A = 32$ by approximately 1.08×10^{-4} at eight substeps, but changes $A = 40$ by approximately 36.8% and suppresses selected sectors from $A = 56$ upward by more than 99.99%. At 64 substeps with the same cutoff, positive support still reaches every interior mass through $A = 159$, while the $A = 119$ magnitude falls to approximately 6.42×10^{-75}.

Accordingly, the outward-boundary and temporal results strengthen the graph-reachability claim, but the deep-tail magnitudes remain sensitive to the earliest and smallest values in the native ^{30}Si trajectory. The calculation rules out silicon-30 as a topological endpoint of this selected reduced graph. It does not establish a finite physical endpoint, a precision-qualified heavy-element yield, or native-network coverage beyond silicon-30.

5 Continuous Nuclide Register from Mass 1 Through Mass 119

For each abundance-bearing nuclide i, Y_i denotes its endpoint number abundance per baryon. The abundance assigned to mass sector A is

$$Y_A = \sum_{i:A_i=A} Y_i. \tag{10}$$

Table 2 contains 119 consecutive mass rows and 1,111 listed nuclide or resonance entries. At fixed mass number, entries belonging to different elements are isobars rather than isotopes; the table therefore uses the more precise term *nuclide*, consistent with IUPAC terminology [6]. Masses 1 through 30 reproduce the complete controlling native endpoint inventory. The ^{8}Be value in that source is the declared 10^{-50}

abundance-floor entry and should not be interpreted as a persistent primordial abundance. Masses 31 through 119 reproduce every node in the one-way prescribed-trajectory neutron-capture and beta-minus graph. A listed diagnostic node records rate-library graph membership; it does not, by itself, establish experimental observation or measurable primordial production. Three listed graph nodes—^{43}Si, ^{47}Si, and ^{51}Si—have exactly zero endpoint abundance in the eight-substep output; every mass-sector sum from $A = 31$ through $A = 119$ remains positive.

The abundance column uses the native endpoint sums for $A = 1$–30 and preserves the archived single $A \leq 120$, eight-substep diagnostic output for $A = 31$–119. Those archived values are retained so that the register remains an exact reproduction object for the original release. Table 1 gives the stronger temporal qualification without silently replacing the archived basis. For D1 and D2 rows, Y_A is a diagnostic endpoint population in the reduced one-way graph and is not asserted to be a physical primordial heavy-element abundance. The column is marked not applicable at $A = 5$ because the listed entries are resonances rather than abundance coordinates. Numerical values are reproduced at the precision stored in the controlling source files; variation in printed trailing digits does not represent an independently assigned uncertainty. The evidence-class codes are defined immediately before the register.

Class codes: N0, native universal light-state core; **N1**, native oxygen-16 basis; **N2**, native post-oxygen extension through silicon-30; **D1**, numerically refined prescribed-trajectory continuation; **D2**, exploratory prescribed-trajectory graph; **R**, listed resonance entries without an abundance coordinate.

Table 2: Gap-free register of the nuclides and resonances represented in the combined calculations for every mass number from $A = 1$ through $A = 119$. The column N gives the number of listed entries at each mass.

A	N	Represented nuclides	Archived Y_A	Class
1	2	n, ^{1}H (p)	7.522677×10^{-1}	N0
2	1	^{2}H (D)	1.860436×10^{-5}	N0
3	2	^{3}H (T), ^{3}He	7.789007×10^{-6}	N0
4	1	^{4}He	6.191793×10^{-2}	N0
5	2	^{5}He, ^{5}Li	not applicable	R
6	1	^{6}Li	8.214136×10^{-15}	N1
7	2	^{7}Li, ^{7}Be	4.024159×10^{-10}	N1
8	3	^{8}Li, ^{8}Be, ^{8}B	2.246811×10^{-34}	N1
9	1	^{9}Be	1.86585×10^{-19}	N1
10	2	^{10}Be, ^{10}B	2.468639×10^{-21}	N1
11	2	^{11}B, ^{11}C	3.495781×10^{-16}	N1

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A	N	Represented nuclides	Archived Y_A	Class
12	3	^{12}B, ^{12}C, ^{12}N	3.582536×10^{-16}	N1
13	2	^{13}C, ^{13}N	7.450479×10^{-17}	N1
14	3	^{14}C, ^{14}N, ^{14}O	4.090225×10^{-17}	N1
15	2	^{15}N, ^{15}O	9.790313×10^{-21}	N1
16	1	^{16}O	3.57075×10^{-20}	N1
17	1	^{17}O	3.134831×10^{-22}	N2
18	2	^{18}O, ^{18}F	1.020704×10^{-24}	N2
19	1	^{19}F	2.790054×10^{-27}	N2
20	1	^{20}Ne	1.298913×10^{-28}	N2
21	1	^{21}Ne	5.006889×10^{-30}	N2
22	2	^{22}Ne, ^{22}Na	6.868353×10^{-30}	N2
23	2	^{23}Ne, ^{23}Na	1.022984×10^{-31}	N2
24	1	^{24}Mg	2.630659×10^{-35}	N2
25	1	^{25}Mg	1.678778×10^{-36}	N2
26	2	^{26}Mg, ^{26}Al	2.364879×10^{-36}	N2
27	1	^{27}Al	5.198167×10^{-39}	N2
28	1	^{28}Si	1.14832×10^{-44}	N2
29	1	^{29}Si	3.693064×10^{-46}	N2
30	1	^{30}Si	1.074185×10^{-46}	N2
31	2	^{31}Si, ^{31}P	1.260807×10^{-47}	D1
32	3	^{32}Si, ^{32}P, ^{32}S	7.165959×10^{-48}	D1
33	3	^{33}Si, ^{33}P, ^{33}S	6.418241×10^{-49}	D2
34	3	^{34}Si, ^{34}P, ^{34}S	2.919824×10^{-49}	D2
35	4	^{35}Si, ^{35}P, ^{35}S, ^{35}Cl	1.435181×10^{-49}	D2

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A	N	Represented nuclides	Archived Y_A	Class
36	5	^{36}Si, ^{36}P, ^{36}S, ^{36}Cl, ^{36}Ar	5.546139×10^{-50}	D2
37	5	^{37}Si, ^{37}P, ^{37}S, ^{37}Cl, ^{37}Ar	5.235818×10^{-51}	D2
38	5	^{38}Si, ^{38}P, ^{38}S, ^{38}Cl, ^{38}Ar	2.698372×10^{-51}	D2
39	6	^{39}Si, ^{39}P, ^{39}S, ^{39}Cl, ^{39}Ar, ^{39}K	4.871949×10^{-52}	D2
40	7	^{40}Si, ^{40}P, ^{40}S, ^{40}Cl, ^{40}Ar, ^{40}K, ^{40}Ca	1.687562×10^{-52}	D2
41	7	^{41}Si, ^{41}P, ^{41}S, ^{41}Cl, ^{41}Ar, ^{41}K, ^{41}Ca	2.217256×10^{-52}	D2
42	7	^{42}Si, ^{42}P, ^{42}S, ^{42}Cl, ^{42}Ar, ^{42}K, ^{42}Ca	1.8769×10^{-52}	D2
43	7	^{43}Si, ^{43}P, ^{43}S, ^{43}Cl, ^{43}Ar, ^{43}K, ^{43}Ca	8.938921×10^{-53}	D2
44	7	^{44}Si, ^{44}P, ^{44}S, ^{44}Cl, ^{44}Ar, ^{44}K, ^{44}Ca	1.271115×10^{-52}	D2
45	8	^{45}Si, ^{45}P, ^{45}S, ^{45}Cl, ^{45}Ar, ^{45}K, ^{45}Ca, ^{45}Sc	5.279289×10^{-53}	D2
46	9	^{46}Si, ^{46}P, ^{46}S, ^{46}Cl, ^{46}Ar, ^{46}K, ^{46}Ca, ^{46}Sc, ^{46}Ti	4.728164×10^{-53}	D2
47	9	^{47}Si, ^{47}P, ^{47}S, ^{47}Cl, ^{47}Ar, ^{47}K, ^{47}Ca, ^{47}Sc, ^{47}Ti	3.905347×10^{-53}	D2
48	9	^{48}Si, ^{48}P, ^{48}S, ^{48}Cl, ^{48}Ar, ^{48}K, ^{48}Ca, ^{48}Sc, ^{48}Ti	3.351341×10^{-53}	D2
49	9	^{49}Si, ^{49}P, ^{49}S, ^{49}Cl, ^{49}Ar, ^{49}K, ^{49}Ca, ^{49}Sc, ^{49}Ti	2.322291×10^{-53}	D2
50	9	^{50}Si, ^{50}P, ^{50}S, ^{50}Cl, ^{50}Ar, ^{50}K, ^{50}Ca, ^{50}Sc, ^{50}Ti	1.119949×10^{-53}	D2
51	10	^{51}Si, ^{51}P, ^{51}S, ^{51}Cl, ^{51}Ar, ^{51}K, ^{51}Ca, ^{51}Sc, ^{51}Ti, ^{51}V	6.471204×10^{-54}	D2
52	9	^{52}S, ^{52}Cl, ^{52}Ar, ^{52}K, ^{52}Ca, ^{52}Sc, ^{52}Ti, ^{52}V, ^{52}Cr	4.588623×10^{-54}	D2
53	9	^{53}S, ^{53}Cl, ^{53}Ar, ^{53}K, ^{53}Ca, ^{53}Sc, ^{53}Ti, ^{53}V, ^{53}Cr	3.067894×10^{-54}	D2
54	9	^{54}S, ^{54}Cl, ^{54}Ar, ^{54}K, ^{54}Ca, ^{54}Sc, ^{54}Ti, ^{54}V, ^{54}Cr	1.037553×10^{-54}	D2
55	10	^{55}S, ^{55}Cl, ^{55}Ar, ^{55}K, ^{55}Ca, ^{55}Sc, ^{55}Ti, ^{55}V, ^{55}Cr, ^{55}Mn	1.169812×10^{-54}	D2
56	11	^{56}S, ^{56}Cl, ^{56}Ar, ^{56}K, ^{56}Ca, ^{56}Sc, ^{56}Ti, ^{56}V, ^{56}Cr, ^{56}Mn, ^{56}Fe	1.765245×10^{-54}	D2
57	11	^{57}S, ^{57}Cl, ^{57}Ar, ^{57}K, ^{57}Ca, ^{57}Sc, ^{57}Ti, ^{57}V, ^{57}Cr, ^{57}Mn, ^{57}Fe	7.957791×10^{-55}	D2
58	11	^{58}S, ^{58}Cl, ^{58}Ar, ^{58}K, ^{58}Ca, ^{58}Sc, ^{58}Ti, ^{58}V, ^{58}Cr, ^{58}Mn, ^{58}Fe	5.671194×10^{-55}	D2
59	11	^{59}Cl, ^{59}Ar, ^{59}K, ^{59}Ca, ^{59}Sc, ^{59}Ti, ^{59}V, ^{59}Cr, ^{59}Mn, ^{59}Fe, ^{59}Co	4.114257×10^{-55}	D2

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A	N	Represented nuclides	Archived Y_A	Class
60	12	^{60}Cl, ^{60}Ar, ^{60}K, ^{60}Ca, ^{60}Sc, ^{60}Ti, ^{60}V, ^{60}Cr, ^{60}Mn, ^{60}Fe, ^{60}Co, ^{60}Ni	9.994849×10^{-55}	D2
61	12	^{61}Cl, ^{61}Ar, ^{61}K, ^{61}Ca, ^{61}Sc, ^{61}Ti, ^{61}V, ^{61}Cr, ^{61}Mn, ^{61}Fe, ^{61}Co, ^{61}Ni	4.205458×10^{-55}	D2
62	11	^{62}Ar, ^{62}K, ^{62}Ca, ^{62}Sc, ^{62}Ti, ^{62}V, ^{62}Cr, ^{62}Mn, ^{62}Fe, ^{62}Co, ^{62}Ni	6.831138×10^{-55}	D2
63	12	^{63}Ar, ^{63}K, ^{63}Ca, ^{63}Sc, ^{63}Ti, ^{63}V, ^{63}Cr, ^{63}Mn, ^{63}Fe, ^{63}Co, ^{63}Ni, ^{63}Cu	3.540761×10^{-55}	D2
64	13	^{64}Ar, ^{64}K, ^{64}Ca, ^{64}Sc, ^{64}Ti, ^{64}V, ^{64}Cr, ^{64}Mn, ^{64}Fe, ^{64}Co, ^{64}Ni, ^{64}Cu, ^{64}Zn	3.755918×10^{-55}	D2
65	12	^{65}K, ^{65}Ca, ^{65}Sc, ^{65}Ti, ^{65}V, ^{65}Cr, ^{65}Mn, ^{65}Fe, ^{65}Co, ^{65}Ni, ^{65}Cu, ^{65}Zn	1.504781×10^{-55}	D2
66	12	^{66}K, ^{66}Ca, ^{66}Sc, ^{66}Ti, ^{66}V, ^{66}Cr, ^{66}Mn, ^{66}Fe, ^{66}Co, ^{66}Ni, ^{66}Cu, ^{66}Zn	4.948077×10^{-55}	D2
67	12	^{67}K, ^{67}Ca, ^{67}Sc, ^{67}Ti, ^{67}V, ^{67}Cr, ^{67}Mn, ^{67}Fe, ^{67}Co, ^{67}Ni, ^{67}Cu, ^{67}Zn	1.517976×10^{-55}	D2
68	11	^{68}Ca, ^{68}Sc, ^{68}Ti, ^{68}V, ^{68}Cr, ^{68}Mn, ^{68}Fe, ^{68}Co, ^{68}Ni, ^{68}Cu, ^{68}Zn	3.464737×10^{-55}	D2
69	12	^{69}Ca, ^{69}Sc, ^{69}Ti, ^{69}V, ^{69}Cr, ^{69}Mn, ^{69}Fe, ^{69}Co, ^{69}Ni, ^{69}Cu, ^{69}Zn, ^{69}Ga	2.851959×10^{-55}	D2
70	13	^{70}Ca, ^{70}Sc, ^{70}Ti, ^{70}V, ^{70}Cr, ^{70}Mn, ^{70}Fe, ^{70}Co, ^{70}Ni, ^{70}Cu, ^{70}Zn, ^{70}Ga, ^{70}Ge	1.086518×10^{-55}	D2
71	13	^{71}Ca, ^{71}Sc, ^{71}Ti, ^{71}V, ^{71}Cr, ^{71}Mn, ^{71}Fe, ^{71}Co, ^{71}Ni, ^{71}Cu, ^{71}Zn, ^{71}Ga, ^{71}Ge	1.460832×10^{-55}	D2
72	12	^{72}Sc, ^{72}Ti, ^{72}V, ^{72}Cr, ^{72}Mn, ^{72}Fe, ^{72}Co, ^{72}Ni, ^{72}Cu, ^{72}Zn, ^{72}Ga, ^{72}Ge	1.863515×10^{-55}	D2
73	12	^{73}Sc, ^{73}Ti, ^{73}V, ^{73}Cr, ^{73}Mn, ^{73}Fe, ^{73}Co, ^{73}Ni, ^{73}Cu, ^{73}Zn, ^{73}Ga, ^{73}Ge	8.537773×10^{-56}	D2
74	12	^{74}Sc, ^{74}Ti, ^{74}V, ^{74}Cr, ^{74}Mn, ^{74}Fe, ^{74}Co, ^{74}Ni, ^{74}Cu, ^{74}Zn, ^{74}Ga, ^{74}Ge	2.819861×10^{-55}	D2
75	12	^{75}Ti, ^{75}V, ^{75}Cr, ^{75}Mn, ^{75}Fe, ^{75}Co, ^{75}Ni, ^{75}Cu, ^{75}Zn, ^{75}Ga, ^{75}Ge, ^{75}As	1.851494×10^{-55}	D2
76	13	^{76}Ti, ^{76}V, ^{76}Cr, ^{76}Mn, ^{76}Fe, ^{76}Co, ^{76}Ni, ^{76}Cu, ^{76}Zn, ^{76}Ga, ^{76}Ge, ^{76}As, ^{76}Se	5.175386×10^{-56}	D2
77	13	^{77}Ti, ^{77}V, ^{77}Cr, ^{77}Mn, ^{77}Fe, ^{77}Co, ^{77}Ni, ^{77}Cu, ^{77}Zn, ^{77}Ga, ^{77}Ge, ^{77}As, ^{77}Se	1.020116×10^{-55}	D2
78	12	^{78}V, ^{78}Cr, ^{78}Mn, ^{78}Fe, ^{78}Co, ^{78}Ni, ^{78}Cu, ^{78}Zn, ^{78}Ga, ^{78}Ge, ^{78}As, ^{78}Se	2.107071×10^{-55}	D2
79	13	^{79}V, ^{79}Cr, ^{79}Mn, ^{79}Fe, ^{79}Co, ^{79}Ni, ^{79}Cu, ^{79}Zn, ^{79}Ga, ^{79}Ge, ^{79}As, ^{79}Se, ^{79}Br	5.182884×10^{-56}	D2
80	14	^{80}V, ^{80}Cr, ^{80}Mn, ^{80}Fe, ^{80}Co, ^{80}Ni, ^{80}Cu, ^{80}Zn, ^{80}Ga, ^{80}Ge, ^{80}As, ^{80}Se, ^{80}Br, ^{80}Kr	1.016253×10^{-55}	D2
81	13	^{81}Cr, ^{81}Mn, ^{81}Fe, ^{81}Co, ^{81}Ni, ^{81}Cu, ^{81}Zn, ^{81}Ga, ^{81}Ge, ^{81}As, ^{81}Se, ^{81}Br, ^{81}Kr	1.210256×10^{-55}	D2
82	13	^{82}Cr, ^{82}Mn, ^{82}Fe, ^{82}Co, ^{82}Ni, ^{82}Cu, ^{82}Zn, ^{82}Ga, ^{82}Ge, ^{82}As, ^{82}Se, ^{82}Br, ^{82}Kr	1.046129×10^{-55}	D2

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A	N	Represented nuclides	Archived Y_A	Class
83	13	^{83}Cr, ^{83}Mn, ^{83}Fe, ^{83}Co, ^{83}Ni, ^{83}Cu, ^{83}Zn, ^{83}Ga, ^{83}Ge, ^{83}As, ^{83}Se, ^{83}Br, ^{83}Kr	1.597622×10^{-55}	D2
84	13	^{84}Cr, ^{84}Mn, ^{84}Fe, ^{84}Co, ^{84}Ni, ^{84}Cu, ^{84}Zn, ^{84}Ga, ^{84}Ge, ^{84}As, ^{84}Se, ^{84}Br, ^{84}Kr	2.743322×10^{-55}	D2
85	13	^{85}Mn, ^{85}Fe, ^{85}Co, ^{85}Ni, ^{85}Cu, ^{85}Zn, ^{85}Ga, ^{85}Ge, ^{85}As, ^{85}Se, ^{85}Br, ^{85}Kr, ^{85}Rb	2.316254×10^{-55}	D2
86	14	^{86}Mn, ^{86}Fe, ^{86}Co, ^{86}Ni, ^{86}Cu, ^{86}Zn, ^{86}Ga, ^{86}Ge, ^{86}As, ^{86}Se, ^{86}Br, ^{86}Kr, ^{86}Rb, ^{86}Sr	1.44283×10^{-55}	D2
87	14	^{87}Mn, ^{87}Fe, ^{87}Co, ^{87}Ni, ^{87}Cu, ^{87}Zn, ^{87}Ga, ^{87}Ge, ^{87}As, ^{87}Se, ^{87}Br, ^{87}Kr, ^{87}Rb, ^{87}Sr	6.053177×10^{-56}	D2
88	13	^{88}Fe, ^{88}Co, ^{88}Ni, ^{88}Cu, ^{88}Zn, ^{88}Ga, ^{88}Ge, ^{88}As, ^{88}Se, ^{88}Br, ^{88}Kr, ^{88}Rb, ^{88}Sr	2.926046×10^{-56}	D2
89	14	^{89}Fe, ^{89}Co, ^{89}Ni, ^{89}Cu, ^{89}Zn, ^{89}Ga, ^{89}Ge, ^{89}As, ^{89}Se, ^{89}Br, ^{89}Kr, ^{89}Rb, ^{89}Sr, ^{89}Y	6.1468×10^{-56}	D2
90	15	^{90}Fe, ^{90}Co, ^{90}Ni, ^{90}Cu, ^{90}Zn, ^{90}Ga, ^{90}Ge, ^{90}As, ^{90}Se, ^{90}Br, ^{90}Kr, ^{90}Rb, ^{90}Sr, ^{90}Y, ^{90}Zr	4.745019×10^{-56}	D2
91	14	^{91}Co, ^{91}Ni, ^{91}Cu, ^{91}Zn, ^{91}Ga, ^{91}Ge, ^{91}As, ^{91}Se, ^{91}Br, ^{91}Kr, ^{91}Rb, ^{91}Sr, ^{91}Y, ^{91}Zr	4.245786×10^{-56}	D2
92	14	^{92}Co, ^{92}Ni, ^{92}Cu, ^{92}Zn, ^{92}Ga, ^{92}Ge, ^{92}As, ^{92}Se, ^{92}Br, ^{92}Kr, ^{92}Rb, ^{92}Sr, ^{92}Y, ^{92}Zr	3.309035×10^{-56}	D2
93	15	^{93}Co, ^{93}Ni, ^{93}Cu, ^{93}Zn, ^{93}Ga, ^{93}Ge, ^{93}As, ^{93}Se, ^{93}Br, ^{93}Kr, ^{93}Rb, ^{93}Sr, ^{93}Y, ^{93}Zr, ^{93}Nb	2.900263×10^{-56}	D2
94	15	^{94}Ni, ^{94}Cu, ^{94}Zn, ^{94}Ga, ^{94}Ge, ^{94}As, ^{94}Se, ^{94}Br, ^{94}Kr, ^{94}Rb, ^{94}Sr, ^{94}Y, ^{94}Zr, ^{94}Nb, ^{94}Mo	3.931671×10^{-56}	D2
95	15	^{95}Ni, ^{95}Cu, ^{95}Zn, ^{95}Ga, ^{95}Ge, ^{95}As, ^{95}Se, ^{95}Br, ^{95}Kr, ^{95}Rb, ^{95}Sr, ^{95}Y, ^{95}Zr, ^{95}Nb, ^{95}Mo	2.978948×10^{-56}	D2
96	15	^{96}Ni, ^{96}Cu, ^{96}Zn, ^{96}Ga, ^{96}Ge, ^{96}As, ^{96}Se, ^{96}Br, ^{96}Kr, ^{96}Rb, ^{96}Sr, ^{96}Y, ^{96}Zr, ^{96}Nb, ^{96}Mo	1.634314×10^{-56}	D2
97	14	^{97}Cu, ^{97}Zn, ^{97}Ga, ^{97}Ge, ^{97}As, ^{97}Se, ^{97}Br, ^{97}Kr, ^{97}Rb, ^{97}Sr, ^{97}Y, ^{97}Zr, ^{97}Nb, ^{97}Mo	2.386422×10^{-56}	D2
98	14	^{98}Cu, ^{98}Zn, ^{98}Ga, ^{98}Ge, ^{98}As, ^{98}Se, ^{98}Br, ^{98}Kr, ^{98}Rb, ^{98}Sr, ^{98}Y, ^{98}Zr, ^{98}Nb, ^{98}Mo	1.19575×10^{-56}	D2
99	16	^{99}Cu, ^{99}Zn, ^{99}Ga, ^{99}Ge, ^{99}As, ^{99}Se, ^{99}Br, ^{99}Kr, ^{99}Rb, ^{99}Sr, ^{99}Y, ^{99}Zr, ^{99}Nb, ^{99}Mo, ^{99}Tc, ^{99}Ru	4.010949×10^{-57}	D2
100	16	^{100}Cu, ^{100}Zn, ^{100}Ga, ^{100}Ge, ^{100}As, ^{100}Se, ^{100}Br, ^{100}Kr, ^{100}Rb, ^{100}Sr, ^{100}Y, ^{100}Zr, ^{100}Nb, ^{100}Mo, ^{100}Tc, ^{100}Ru	4.66066×10^{-57}	D2
101	15	^{101}Zn, ^{101}Ga, ^{101}Ge, ^{101}As, ^{101}Se, ^{101}Br, ^{101}Kr, ^{101}Rb, ^{101}Sr, ^{101}Y, ^{101}Zr, ^{101}Nb, ^{101}Mo, ^{101}Tc, ^{101}Ru	3.713053×10^{-57}	D2

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A	N	Represented nuclides	Archived Y_A	Class
102	15	^{102}Zn, ^{102}Ga, ^{102}Ge, ^{102}As, ^{102}Se, ^{102}Br, ^{102}Kr, ^{102}Rb, ^{102}Sr, ^{102}Y, ^{102}Zr, ^{102}Nb, ^{102}Mo, ^{102}Tc, ^{102}Ru	8.071407×10^{-57}	D2
103	16	^{103}Zn, ^{103}Ga, ^{103}Ge, ^{103}As, ^{103}Se, ^{103}Br, ^{103}Kr, ^{103}Rb, ^{103}Sr, ^{103}Y, ^{103}Zr, ^{103}Nb, ^{103}Mo, ^{103}Tc, ^{103}Ru, ^{103}Rh	2.487525×10^{-57}	D2
104	16	^{104}Ga, ^{104}Ge, ^{104}As, ^{104}Se, ^{104}Br, ^{104}Kr, ^{104}Rb, ^{104}Sr, ^{104}Y, ^{104}Zr, ^{104}Nb, ^{104}Mo, ^{104}Tc, ^{104}Ru, ^{104}Rh, ^{104}Pd	6.492919×10^{-57}	D2
105	16	^{105}Ga, ^{105}Ge, ^{105}As, ^{105}Se, ^{105}Br, ^{105}Kr, ^{105}Rb, ^{105}Sr, ^{105}Y, ^{105}Zr, ^{105}Nb, ^{105}Mo, ^{105}Tc, ^{105}Ru, ^{105}Rh, ^{105}Pd	8.021328×10^{-57}	D2
106	16	^{106}Ga, ^{106}Ge, ^{106}As, ^{106}Se, ^{106}Br, ^{106}Kr, ^{106}Rb, ^{106}Sr, ^{106}Y, ^{106}Zr, ^{106}Nb, ^{106}Mo, ^{106}Tc, ^{106}Ru, ^{106}Rh, ^{106}Pd	3.052984×10^{-57}	D2
107	16	^{107}Ge, ^{107}As, ^{107}Se, ^{107}Br, ^{107}Kr, ^{107}Rb, ^{107}Sr, ^{107}Y, ^{107}Zr, ^{107}Nb, ^{107}Mo, ^{107}Tc, ^{107}Ru, ^{107}Rh, ^{107}Pd, ^{107}Ag	4.27056×10^{-57}	D2
108	17	^{108}Ge, ^{108}As, ^{108}Se, ^{108}Br, ^{108}Kr, ^{108}Rb, ^{108}Sr, ^{108}Y, ^{108}Zr, ^{108}Nb, ^{108}Mo, ^{108}Tc, ^{108}Ru, ^{108}Rh, ^{108}Pd, ^{108}Ag, ^{108}Cd	6.845493×10^{-57}	D2
109	17	^{109}Ge, ^{109}As, ^{109}Se, ^{109}Br, ^{109}Kr, ^{109}Rb, ^{109}Sr, ^{109}Y, ^{109}Zr, ^{109}Nb, ^{109}Mo, ^{109}Tc, ^{109}Ru, ^{109}Rh, ^{109}Pd, ^{109}Ag, ^{109}Cd	2.022383×10^{-57}	D2
110	16	^{110}As, ^{110}Se, ^{110}Br, ^{110}Kr, ^{110}Rb, ^{110}Sr, ^{110}Y, ^{110}Zr, ^{110}Nb, ^{110}Mo, ^{110}Tc, ^{110}Ru, ^{110}Rh, ^{110}Pd, ^{110}Ag, ^{110}Cd	5.998031×10^{-57}	D2
111	16	^{111}As, ^{111}Se, ^{111}Br, ^{111}Kr, ^{111}Rb, ^{111}Sr, ^{111}Y, ^{111}Zr, ^{111}Nb, ^{111}Mo, ^{111}Tc, ^{111}Ru, ^{111}Rh, ^{111}Pd, ^{111}Ag, ^{111}Cd	2.259657×10^{-57}	D2
112	16	^{112}As, ^{112}Se, ^{112}Br, ^{112}Kr, ^{112}Rb, ^{112}Sr, ^{112}Y, ^{112}Zr, ^{112}Nb, ^{112}Mo, ^{112}Tc, ^{112}Ru, ^{112}Rh, ^{112}Pd, ^{112}Ag, ^{112}Cd	6.772206×10^{-57}	D2
113	16	^{113}Se, ^{113}Br, ^{113}Kr, ^{113}Rb, ^{113}Sr, ^{113}Y, ^{113}Zr, ^{113}Nb, ^{113}Mo, ^{113}Tc, ^{113}Ru, ^{113}Rh, ^{113}Pd, ^{113}Ag, ^{113}Cd, ^{113}In	1.479664×10^{-57}	D2
114	17	^{114}Se, ^{114}Br, ^{114}Kr, ^{114}Rb, ^{114}Sr, ^{114}Y, ^{114}Zr, ^{114}Nb, ^{114}Mo, ^{114}Tc, ^{114}Ru, ^{114}Rh, ^{114}Pd, ^{114}Ag, ^{114}Cd, ^{114}In, ^{114}Sn	7.965147×10^{-57}	D2
115	17	^{115}Se, ^{115}Br, ^{115}Kr, ^{115}Rb, ^{115}Sr, ^{115}Y, ^{115}Zr, ^{115}Nb, ^{115}Mo, ^{115}Tc, ^{115}Ru, ^{115}Rh, ^{115}Pd, ^{115}Ag, ^{115}Cd, ^{115}In, ^{115}Sn	3.090037×10^{-57}	D2
116	17	^{116}Se, ^{116}Br, ^{116}Kr, ^{116}Rb, ^{116}Sr, ^{116}Y, ^{116}Zr, ^{116}Nb, ^{116}Mo, ^{116}Tc, ^{116}Ru, ^{116}Rh, ^{116}Pd, ^{116}Ag, ^{116}Cd, ^{116}In, ^{116}Sn	2.0995×10^{-57}	D2
117	16	^{117}Br, ^{117}Kr, ^{117}Rb, ^{117}Sr, ^{117}Y, ^{117}Zr, ^{117}Nb, ^{117}Mo, ^{117}Tc, ^{117}Ru, ^{117}Rh, ^{117}Pd, ^{117}Ag, ^{117}Cd, ^{117}In, ^{117}Sn	5.006565×10^{-57}	D2
118	16	^{118}Br, ^{118}Kr, ^{118}Rb, ^{118}Sr, ^{118}Y, ^{118}Zr, ^{118}Nb, ^{118}Mo, ^{118}Tc, ^{118}Ru, ^{118}Rh, ^{118}Pd, ^{118}Ag, ^{118}Cd, ^{118}In, ^{118}Sn	7.760345×10^{-57}	D2
119	16	^{119}Br, ^{119}Kr, ^{119}Rb, ^{119}Sr, ^{119}Y, ^{119}Zr, ^{119}Nb, ^{119}Mo, ^{119}Tc, ^{119}Ru, ^{119}Rh, ^{119}Pd, ^{119}Ag, ^{119}Cd, ^{119}In, ^{119}Sn	2.873667×10^{-57}	D2

6 The Mass-Five Gap

Mass five is present in the continuous register so that the data contain no missing mass row. The two mass-five resonances explicitly represented in this construction,

$$^{5}\mathrm{He} \qquad \text{and} \qquad {}^{5}\mathrm{Li}, \tag{11}$$

are unbound resonances:

$$^{5}\mathrm{He} \longrightarrow {}^{4}\mathrm{He} + n, \tag{12}$$
$$^{5}\mathrm{Li} \longrightarrow {}^{4}\mathrm{He} + p. \tag{13}$$

They are broad, particle-unstable resonances and do not accumulate as persistent abundance-bearing states under the conditions represented here. The row is not an exhaustive catalog of every evaluated $A = 5$ system. It records the two resonances relevant to the represented $n + {}^{4}\mathrm{He}$ and $p + {}^{4}\mathrm{He}$ channels and assigns neither an endpoint abundance nor an independent native coordinate. The network can bypass the gap through channels such as

$$^{2}\mathrm{H} + {}^{4}\mathrm{He} \longrightarrow {}^{6}\mathrm{Li} + \gamma, \tag{14}$$
$$^{3}\mathrm{H} + {}^{4}\mathrm{He} \longrightarrow {}^{7}\mathrm{Li} + \gamma, \tag{15}$$
$$^{3}\mathrm{He} + {}^{4}\mathrm{He} \longrightarrow {}^{7}\mathrm{Be} + \gamma. \tag{16}$$

The mass-five row is consequently a physical gap designation, not a missing-data entry.

7 Completeness and Support Classification

The continuous register satisfies the following checks:

$$N_{\text{mass rows}} = 119, \tag{17}$$
$$N_{\text{missing mass rows}} = 0, \tag{18}$$
$$N_{\text{listed nuclide/resonance entries}} = 1111, \tag{19}$$
$$N_{\text{zero endpoint graph nodes}} = 3, \tag{20}$$
$$A_{\text{min}} = 1, \tag{21}$$
$$A_{\text{max}} = 119. \tag{22}$$

The evidence classification is

$$A = 1\text{–}30 : \text{native inventory}, \tag{23}$$
$$A = 31\text{–}32 : \text{strong prescribed-trajectory diagnostic}, \tag{24}$$
$$A = 33\text{–}119 : \text{exploratory prescribed-trajectory graph}. \tag{25}$$

The nuclide lists at masses 31 through 119 describe membership in the reaction graph. They are not claims that all listed nuclides obtain measurable primordial abundances, nor are they a catalog of every experimentally known nuclide at those masses. Some ReacLib graph nodes are supported by evaluated or theoretical rates; graph membership alone is not an experimental-discovery classification. The companion nuclide-level CSV preserves the same register in machine-readable form, including A, nuclide name, proton number, endpoint abundance when applicable, evidence class, and the explicit mass-five resonance note. The three zero-endpoint graph nodes are retained because they are members of the generated reaction graph even though their final floating-point abundance is zero.

8 Scientific Boundary

The boundaries in this section constrain the evidentiary level of particular numerical claims; they do not diminish the unprecedented cross-scale significance established above. The complete register demonstrates continuity of the represented calculation but does not erase the distinction between native-network and diagnostic support. Native-network coverage presently ends at silicon-30. The first post-silicon block is temporally well controlled on the prescribed trajectory. The deeper tail remains an exploratory reduced-graph result.

The new qualification removes the earlier ambiguity about the $A = 120$ boundary: moving the boundary to $A = 140$ and $A = 160$ leaves all 89 mass-sector sums from $A = 31$ through $A = 119$ exactly unchanged. The qualification also establishes that downstream states begin at exact zero and vanish identically when the ^{30}Si source is removed. However, source-cutoff tests show that the deep-tail magnitudes depend strongly on the early sub-10^{-50} portion of that native source trajectory. This dependence does not erase graph reachability, but it prevents the smallest tabulated values from being interpreted as precision-qualified physical abundances.

Accordingly, the current result is:

> The combined native and prescribed-trajectory calculation contains an explicit, gap-free archived mass register from $A = 1$ through $A = 119$. In an enlarged $A_{\max} = 160$ graph, positive diagnostic support persists through every interior mass to $A = 159$, and the original $A = 31$–119 results are exactly independent of the $A = 120$ boundary. This is a reduced-graph reachability result. It does not identify a finite physical endpoint or establish measurable primordial heavy-element yields.

9 Conclusion

This addendum provides a formula-protected evidence companion to the universal CPTG nuclear-reaction theory. It preserves the complete native and diagnostic mass register, documents the exact numerical basis of the post-silicon continuation, and adds source-isolation, source-cutoff, temporal-convergence, and outward-boundary tests.

Its primary scientific conclusion is broader than the location of a numerical frontier. A geometric theory organized by a structural parameter developed for macroscopic gravitating systems has produced a coherent nuclear-reaction architecture from the free-nucleon vertex through the universal light-state sequence and the native inventory to silicon-30. The same native output then seeds a causally controlled continuation whose positive support remains present through every interior mass to $A = 159$. The surveyed nuclear-reaction literature contains many nuclear-internal geometric and macroscopic models, but no identified precedent for this cross-scale identity of the organizing structural parameter.

The strongest new numerical conclusion is that $A = 119$ is not an artifact of placing an absorbing boundary at $A = 120$: the results through $A = 119$ are exactly unchanged when the graph is extended to $A = 140$ and $A = 160$. Positive support also persists at 64-substep resolution and after removing early ^{30}Si source values below 10^{-50}. The magnitude of the deep tail is nevertheless highly source-sensitive.

Taken together, the results establish an unprecedented cross-scale geometric continuation and a robust post-silicon reachability result. They do not yet establish precision-qualified heavy-element yields, a finite physical endpoint, or final independent confirmation of the protected universal law. Those are the next validation thresholds. They are not reasons to describe the present result as routine: the demonstrated survival of one macroscopic structural organization into nuclear-reaction physics is itself the central discovery recorded by this addendum.

10 Data and Code Availability

The controlling release bundle is:
`CPTG_Post016_Si30Bridge_AbsoluteLimitDiagnostic_20260716_r01.zip`

with SHA-256:
`76ee1c5c637069972c231cfda721afbcae1fb8975ca6a128ac2bea5abf79410a`

The extended qualification bundle is:
`CPTG_Post016_Si30FloorBoundaryConvergenceQualification_20260719_r01.zip`

with SHA-256:
`908642caa9dbac8557faba937abcb9d1e488e3cf1d7debb787db27aa3df2c60e`

Within the CPTG repository, the repository-relative release location is:
`https://github.com/CLG2025/CPTG/tree/main/nuclear-reactions`

The principal machine-readable inputs and implementation files are:

- `RESULTS/TRACE_DIAGNOSTIC/raw_endpoint.csv`
  SHA-256 `9e234e2d8f672ad77fbfa1faa26c7fe20020e62325f07eabaa2ae5fdae1272e7`
- `RESULTS/TRACE_DIAGNOSTIC/SI30_NEUTRON_BETA_A120_S8_TRI_ENDPOINT.csv`
  SHA-256 `aecc87d9a58b5e16f2977c2857ddcd4045fd56cf5172ea397b90d1808cda8180`
- `RESULTS/TRACE_DIAGNOSTIC/SI30_NEUTRON_BETA_A120_S8_TRI_BY_MASS.csv`
  SHA-256 `a96b8914c977766e981e017a0054c40b52935a6fa5ab1a2aaf04a3567a797817`
- `SOURCE/si30_neutron_beta_triangular_fast.py`
  SHA-256 `9adb230db299020e7be09fc4e5308a8af70814939242fd9b762e50ec0f73771c`
- `SOURCE/si30_floor_boundary_convergence_qualification.py`
  SHA-256 `99cb37898e65698c482c90feaea64cfa3928c95fa1a6cafb9b40b96ee694c89b`
- `RESULTS/QUALIFICATION_SUMMARY.json`
  source, boundary, convergence, and cutoff decision record

The upstream release identities retained in the bundle are:

- `CPTG_Post016_TheoryExtension_AI_Handoff_20260716_r01.zip`
  SHA-256 `20cbaca5676f9d06dd1cea2be6b45c9c1ecd60dfc07b75032cdc1221b2330c4a`
- `CPTG_Post016_RobustFrontierConfirmation_20260716_r02.zip`
  SHA-256 `75a1855ebf7e3460095ace4383d9c704d161a5f5a7a01cc75f1184f6a3688662`
- `CPTG_Ne23_BBN_vs_ExplosiveRegime_Discrimination_20260716_r01.zip`
  SHA-256 `8757fa29971af6f6375978ee6731b6d04c425e0916bfafc5c6a4de512ef10b60`

The native trajectory was produced with AlterBBN v2.2 at $\eta_{10} = 6.122$, `failsafe`=6, and abundance floor 10^{-50}. The archived prescribed-trajectory continuation used Python 3.13.5, `pynucastro` 2.11.0, a maximum network mass of $A = 120$, and background-interpolated recursive backward Euler. The qualification used the same Python and `pynucastro` versions, extended the graph to $A = 160$, and applied the identical recurrence with Numba 0.65.1 acceleration. No protected CPTG formula or coefficient is present in either the qualification source or its result package.

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